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# Diminished sialoside binding in novel H5N1 influenza hemagglutinin variants identified in a human patient

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In 2024, an adolescent female in British Columbia was hospitalised presenting with severe symptoms including respiratory failure due to infection with a novel H5N1 influenza strain (BC24). Using cryogenic electron microscopy, we show here that the N169 α2,3-linked auto-glycan that is found in the sialic acid binding site of previously studied H5 hemagglutinin (HA) proteins is absent in purified BC24 HA protein, suggesting greatly reduced affinity for α2,3-linked sialosides. Glycan microarray and enzyme-linked immunosorbent assay analyses show that HA variants identified in the BC24 case display severely reduced or no binding to both α2,3-linked sialosides and α2,6-linked sialosides. Full-length BC24 HA expressed in A549 alveolar carcinoma cells drives membrane fusion, albeit at lower levels than previous H5 HA proteins, and post-infection sera from the patient display strong binding to BC24 HA and HA proteins from other influenza subtypes. As each of the two mutations of interest, independently and in conjunction, severely reduce sialoside binding, there appears to have been in this case multiple populations of virus with the diminished receptor binding phenotype. The substantial minority prevalence of weakly binding HA variants in this BC24 case reveals further complexity in the factors that may be present in severe avian influenza infection.

Outbreaks of H5N1 influenza in cattle and birds are a source of concern for zoonotic transmission to humans, with a nearly 50% fatality rate in confirmed cases[1,2]. In November 2024, an adolescent female in British Columbia was hospitalised presenting with severe symptoms including respiratory failure and positivity for infection with novel H5N1 influenza in clade 2.3.4.4b (A/British Columbia/PHL-2032/2024)[3], abbreviated as BC24 in this work. Intact virus was not isolated from the patient, so sequencing information revealed a mixed population of viruses from tracheal aspirate (lower respiratory tract) eight days post-onset of symptoms[3]. In addition to the well-studied E627K mutation in polymerase basic 2, the most frequent mutations observed in H5N1 sequences from this patient were E190D and Q226H (by H3 sequence numbering) in hemagglutinin (HA), each representing approximately a third of sequence reads when samples were taken and thus each demonstrating fitness in this patient[3].

HA is the surface glycoprotein necessary for influenza A infectivity, responsible for both binding host cell receptors and mediating fusion of the viral and host endosomal membranes following endocytosis of the infecting virion[4,5]. The host receptors for HA are typically sialosides: those with α2,3-linked sialic acids (*N*-acetylneuraminic acid (Neu5Ac) in this study) are often considered avian, and those with α2,6 linkages are often considered human[4]. Both are present, however, in human airways, with the α2,6-linked sialosides tending to be enriched in the human upper respiratory epithelia, and α2,3-linked sialosides

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deeper in the respiratory system and in children[4]. The E190D mutation, in concert with G225D, was previously reported to be involved in H1 HA receptor switching to α2,6-linked sialoside binding and human infectivity[6,7]. The E190D mutation in isolation, however, does not appear to promote sialoside affinity in either H1 or H5 contexts[6–8]. Leucine at the 226 position in H5 HA has demonstrated α2,6-linked sialoside binding[5,8]; however, this effect was not found with histidine at this position[5].

To explore the role of these HA mutations observed in BC24 on structure, we carried out cryogenic electron microscopy (cryo-EM) of BC24 HA as well as the H5 HA proteins present in the commonly studied clade 1 A/Vietnam/1203/2004 (VN04) and the mild-human-disease-causing clade 2.3.4.4b A/Michigan/90/2024 (MI24). The latter strain is more recent and closely related to the sequences observed in the infected BC24 patient (sequence alignment provided in Supplementary Fig. 1). Both have been previously described in the literature to bind α2,3-linked sialosides, as is typical for avian influenza HA[7–9]. We specifically assessed the BC24 HA carrying the sequence deposited in GenBank XTJ61994.1 with the two most commonly observed mutations, E190D and Q226H[3]. Sequencing revealed a mixed viral population in the patient, with these mutations representing 28% and 35% of sequence reads, respectively[3]. Therefore, we expect at least a portion of the population to carry Q226H without E190D, and it is unknown to what extent both mutations reside in the same allele[3].

To better understand the glycan-binding properties of these HA proteins, we assessed the binding of BC24 HA to a wide range of glycans in a microarray format[10,11]. We also used enzyme-linked immunosorbent assays (ELISA) to determine the receptor binding effects of the individual E190D and Q226H mutations, noting that receptor binding changes have been observed to arise from a complex interplay of mutations[12–14], and we used fusion assays in A549 human alveolar epithelial carcinoma cells to evaluate the functional effects of the two mutations on HA-mediated membrane fusion under conditions that better mimic the environment during infection.

## Results

### Density for the N169 α2,3-linked auto-glycan is absent in BC24 HA

To determine whether the HA mutations in BC24 result in altered HA conformation, we determined the pre-fusion state cryo-EM structures of trimeric HA ectodomains for VN04, MI24, and BC24 expressed and purified under identical conditions (processing workflow in Supplementary Figs. 2–4). Globally, all HA proteins display similar tertiary and quaternary structures (Figs. 1a–c), with root mean square deviations (RMSD) of BC24 versus VN04 HA being 0.441 Å and of BC24 versus MI24 HA being 1.095 Å. Consistent with preparations at pH 8, we observed well-resolved fusion peptides sequestered in the helix C bundle (Supplementary Fig. 5). The structures demonstrate that the E190D and Q226H mutations do not impart substantial structural changes either close to (Fig. 1d–f) or distal to (Fig. 1a–c) these main residues mutated in BC24 HA.

The most striking finding from comparison of the different H5 HA structures is that clear cryo-EM density for the previously reported N169 auto-glycan[15,16] is observed in the receptor-binding site for both the VN04 and MI24 HA proteins, but not in BC24 HA carrying the E190D and Q226H mutations of interest (Fig. 1d-f). We modelled this full glycan as GlcNAc-GlcNAc-Man-Man-GlcNAc-Gal-Sia as previously reported by Li et al. and Morano et al. for H5 HA proteins in clade

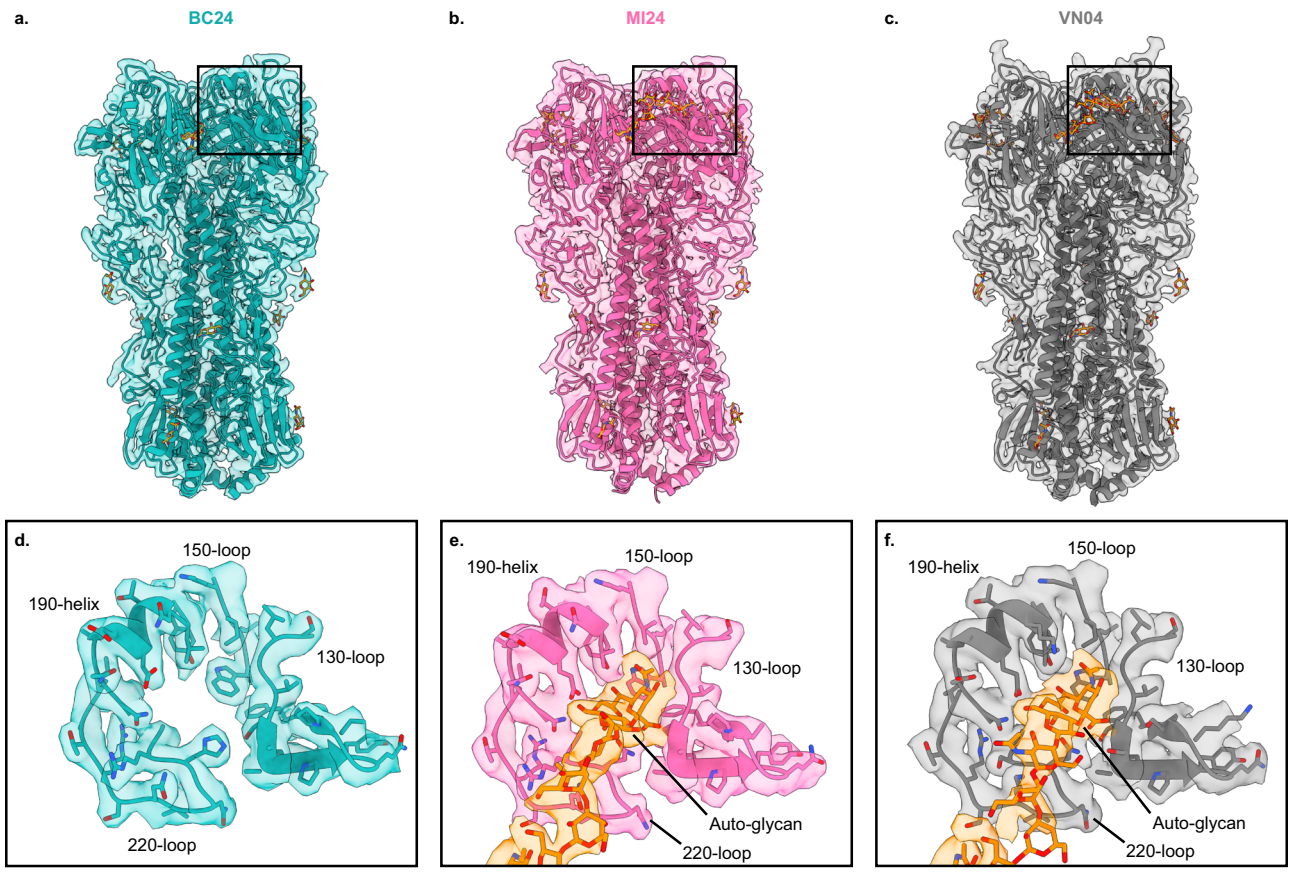



**Fig. 1 | The BC24 HA receptor-binding site lacks auto-glycan density.** Cryo-EM structures (map and model) for HA ectodomain constructs: **a** BC24 (teal), **b** MI24 (pink), and **c** VN04 (grey) HA. Glycosylation in all models is shown in orange. In each case, the black rectangle indicates the region shown in panels (**d**–**f**). To demonstrate loss of the α2,3-linked sialoside auto-glycan binding in the BC24 HA, maps and models of the receptor-binding sites are presented for **d** BC24, **e** MI24, and **f** VN04 H5 HA proteins, with the auto-glycan in orange in each case.

2.3.4.4b expressed in HEK293F cells (Fig. 1e, f)[15,16]. This glycan is likely branched[17], but the branch not bound by the receptor-binding site remains too flexible to resolve. In the normal course of events, these auto-glycans are displaced by sialosides present on the host cell membrane for infection. Asparagine at the 169 position is highly conserved across H5 HA proteins[15], including in the BC24 and VN04 HA proteins. The density maps for VN04 and MI24 HA permit confident modelling of the first two sialic acid and galactose monomers (Sia-1 and Gal-2), showing the terminal glycosidic linkage being in the α2,3 configuration (Fig. 1e, f), distinct from the geometry of the α2,6-linkage[8]. The finding that the BC24 HA receptor-binding site is not normally occupied by the auto-glycan suggests that this HA is likely to have reduced affinity for binding α2,3-linked sialosides on the host cell to initiate viral entry.

## Comparison of glycosylation patterns in BC24 and other H5 HA proteins

HA is known to be extensively glycosylated at multiple sites[18]. We compared the densities for *N*-linked glycans between the different HA proteins to assess whether there were any differences in glycosylation patterns, and to determine whether the absence of auto-glycan in the receptor-binding site is due to a lack of glycosylation at N169 in the BC24 HA protein. The cryo-EM reconstructions show density in the three HA proteins extending from *N*-linked glycosylation sites at positions N21, N33, N158 (VN04 only), N169, N289, and N483 (Fig. 2a–f). We were able to model the first portion at N33, N169, N289, and N483 with sufficient resolution (Fig. 2b, d–f). For N169, density beyond the initial portion is resolved only with HA from VN04 and MI24, where these glycans are bound at the receptor-binding site (Fig. 2d). Although human cells may diversely glycosylate the same site[19], use of the same expression system and the sequence similarity between MI24 and BC24 HA ectodomains suggests that a similar proportion of BC24 HA should be conjugated to an α2,3-linked sialoside at N169 as MI24 HA. The lack of density in the BC24 receptor-binding site (Fig. 1d) is therefore not likely due to a lack of complex glycosylation at N169, but because of reduced affinity for glycans at this position to the receptor-binding site, although it is possible that mutations giving rise to a lack of auto-glycan binding may not play a role in productive binding to receptors. Density connecting glycans to two other residues (N21 and N158, the latter only in VN04) is also visible, but the glycans themselves are likely disordered and could not be modelled reliably (Fig. 2a, c). The presence of the glycan at N158 had previously been reported to moderately mask the receptor-binding site, affecting receptor tropism, and the clade 2.3.4.4b HA sequences lack this glycosylation site, potentially contributing to human infectivity[20].

## Structural explanation for the lack of bound α2,3-linked auto-glycan in BC24 HA

Alignment of the receptor-binding site residues also allows us to compare the bound and unbound structures across these differing H5 HA proteins. The receptor-binding site structure is relatively unchanged in the three H5 HA proteins (Fig. 2g, h), with RMSD values comparing receptor-binding site residues 98, 133-140, 153-156, 183, 187-195, 220-228 being 0.448 Å between VN04 HA and BC24 HA and 0.378 Å between MI24 HA and BC24 HA. Therefore, although the HA structures, in particular MI24 HA, exhibit some preferential orientation, we provide model-based measurements to describe potential differences in α2,3-linked sialoside binding between the three studied HA proteins. In the VN04 and MI24 HA structures, the E190 carboxylate is positioned for hydrogen bonding 2.9 and 2.8 Å, respectively, from the Sia-1 C9 hydroxyl. Their Q226 amides are respectively 3.3 and 3.2 Å from the Gal-2 C4 hydroxyl, and the MI24 Q226 amide is also 3.2 Å from the glycosidic bond oxygen linking Gal-2 and Sia-1. In the absence of bound glycan, the BC24 D190 carboxylate, spaced from the backbone by one fewer methylene moieties, is positioned 4.9 Å from where the MI24-bound Sia-1 aligns. Similarly, H226 of BC24 is pivoted 1.1 Å away from the position of the MI24-bound Gal-2 C4 hydroxyl and 0.9 Å towards the C1 carboxylate, likely permitted by the lack of ligand in the pocket. Thus, key residues in HA that appear well-suited in VN04 and MI24 HA to interact favourably with the α2,3-sialylated N169 auto-glycan are also the ones mutated in BC24 HA, suggesting a structural explanation for the observed lack of α2,3-linked glycan density in the sialic acid binding pocket.

## Glycan microarray analysis demonstrates lack of BC24 HA binding to sialosides

With structural evidence that the BC24 HA may have significantly reduced α2,3-linked sialoside binding, we sought to understand the glycan binding profiles of VN04, MI24, and BC24 HA proteins through microarray analysis featuring a range of glycans (Figs. 3 and 4)[10,11]. To the limit of detection for this technique, the VN04 and MI24 HA proteins demonstrated binding exclusively to α2,3-linked sialosides in both *N*-linked and *O*-linked glycan arrays (Figs. 3a, b and 4a, b).

The MI24 HA had a mild binding preference for sialyl Lewis$^X$ (Figs. 3b and 4b), which is consistent with previous findings[9,21,22]. Several groups have reported that the VN04 HA did not bind to most fucosylated glycans (such as those with sialyl Lewis$^X$ and sialyl Lewis$^A$), whereas the MI24 HA or other clade 2.3.4.4 HA proteins bound well to these glycans, and this expanded breadth of MI24 HA binding is reproduced in our results (Figs. 3a, b and 4a, b)[9,21,22]. A weak but noticeable signal was observed for both of the clade 2.3.4.4b MI24 and BC24 HA proteins to the *N*-linked core $Man_3GlcNAc_2$ glycan (N71 in Fig. 3b, c).

Consistent with the cryo-EM structural findings (Fig. 1), the BC24 HA did not demonstrate binding at the sensitivity of this method to any of the α2,3-linked sialosides or any of the glycans represented on the microarrays (Figs. 3c and4c)[10,11]. We performed dynamic light scattering (DLS) showing that all protein preparations used in the microarray analysis are comparable in stability (Supplementary Fig. 6), which indicates that the differences in binding profiles are intrinsic to the receptor selectivity of the HA proteins analyzed.

## Assessment of glycan binding by E190D and Q226H HA mutants using ELISA experiments

With the reported mixed population of these receptor-binding site mutations in sequencing[3], we generated single and double mutants at positions 190 and 226 to investigate their roles in receptor binding in isolation from each other. In particular, we generated forward E190D and Q226H mutations on MI24 background and reverse D190E and H226Q mutations on BC24 background. The reverse mutations allow us to capture all potential HA variants detected from sequencing in the BC24 case to address the mixed population issue[3].

Using ELISA measurements, we subsequently assessed binding of these HA mutants to typical α2,3-linked sialosides 3SLN-N (represented on the microarray as N1, leftmost bar in Fig. 3) and $3SLN_3$-N and typical α2,6-linked sialosides 6SLN-N and $6SLN_3$-N, noting that previous studies have identified that longer sugars improve HA binding affinity[8]. Consistent with the microarray findings, we observed concentration-dependent binding of both VN04 and MI24 HA to the α2,3-linked sialosides but not to the α2,6-linked sialosides (Fig. 5a, b). Binding to the longer glycan was stronger for both of these HA proteins (Fig. 5a, b). Neither the BC24 HA (Fig. 5i) nor any of the mutants carrying D190 or H226 (Fig. 5c–e, g, h) showed significant binding to either sialoside variety within the tested range up to 50 μg/mL, although D190 may impact α2,3-linked sialoside binding less (Fig. 5c, g). Restoring E190 and Q226 on the BC24 background reverted binding to α2,3-linked sialosides (Fig. 5f). Taken together, both the E190D and Q226H mutations individually reproduce the BC24 HA behaviour of strongly reduced binding to both α2,3-linked and α2,6-linked sialosides, and these mutations appear to be the specific cause of this effect

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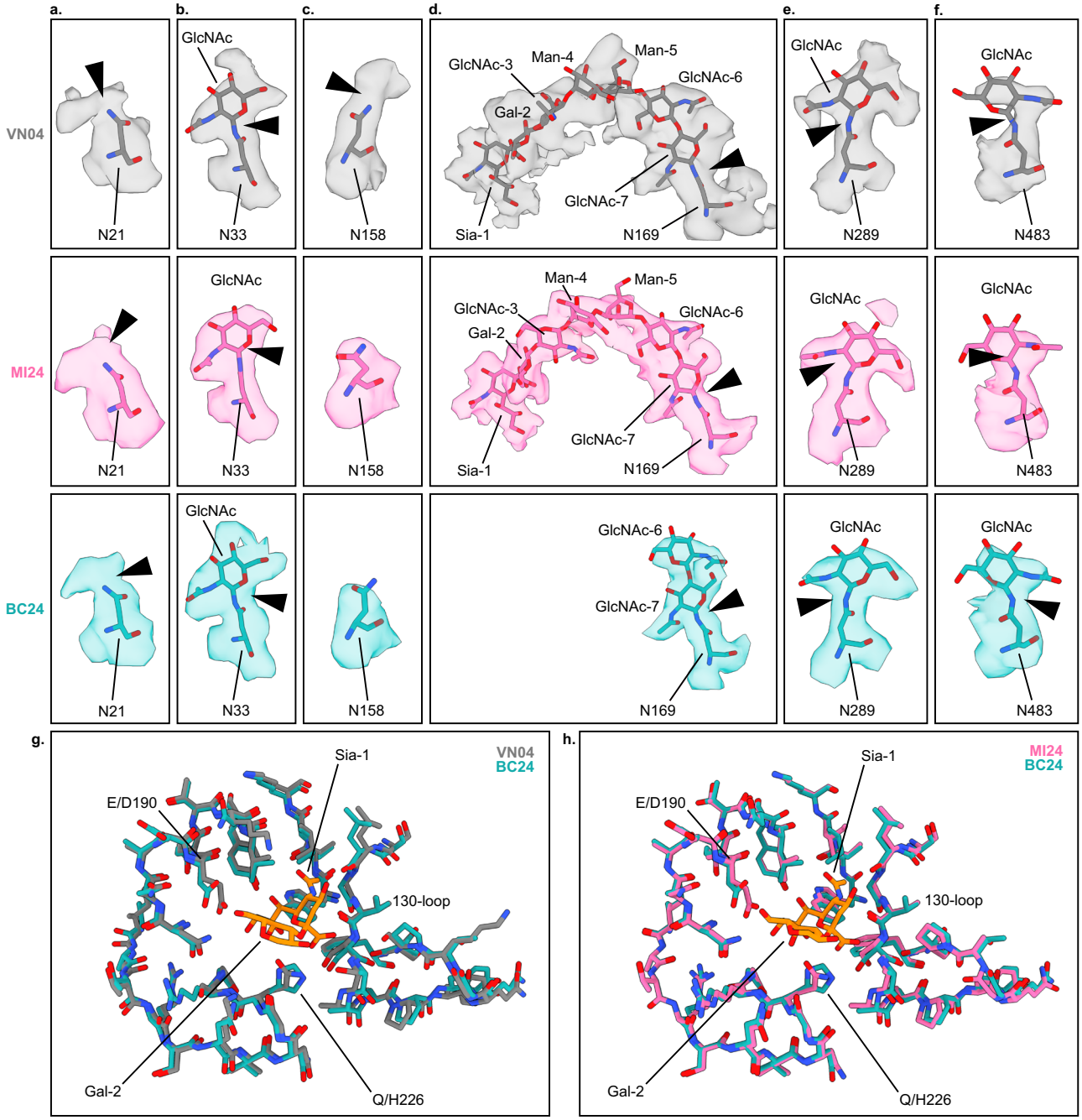



**Fig. 2 | Glycosylation present in H5 HA proteins.** Zoomed-in views of cryo-EM structures (map and model) at the following *N*-glycosylation sites: **a** N21, **b** N33, **c** N158, **d** N169, **e** N289, and **f** N483. Top, VN04 HA; middle, MI24 HA; and bottom, BC24 HA. Arrowheads indicate junctions between glycan and protein density. Sia, sialic acid (Neu5Ac); Gal, galactose; GlcNAc, *N*-acetyl glucosamine; and Man, mannose. Locally aligned cryo-EM models of receptor-binding sites are presented for (**g**). VN04 and BC24 HA proteins and **h**. MI24 and BC24 HA proteins. Teal, BC24; grey, VN04; and pink, MI24. The auto-glycan (in orange) from VN04 and MI24 structures is shown with only the first two residues for clarity.

with differences not being observed between the MI24 or BC24 background.

## BC24 HA is competent for membrane fusion

Since BC24 HA permitted infection of the patient despite the apparently reduced affinity to the assayed glycans, we investigated HA-driven membrane fusion activity in A549 alveolar carcinoma cells transfected with full-length VN04, MI24, or BC24 HA proteins. The transfected cells display HA on the plasma membrane and can be activated to trigger cell-cell fusion by transiently lowering the extracellular pH to mimic the low pH environment of late endosomal membranes. Membrane fusion generates multinucleate syncytia and mixing of the cytoplasm of cell pairs, permitting reconstitution of functional luciferase from cells separately transfected with fluorescently tagged NanoBiT subunits[23,24]. The subsequent luminescence therefore indicates the amount of fusion that occurred. We performed western blotting to verify that variations in luminescence were not due to differential HA expression (Supplementary Fig. 7).

In agreement with the published evidence for BC24 HA function in the case report[3], fusion events are clearly observed in A549 cells transfected with all of the HA proteins, including BC24, by

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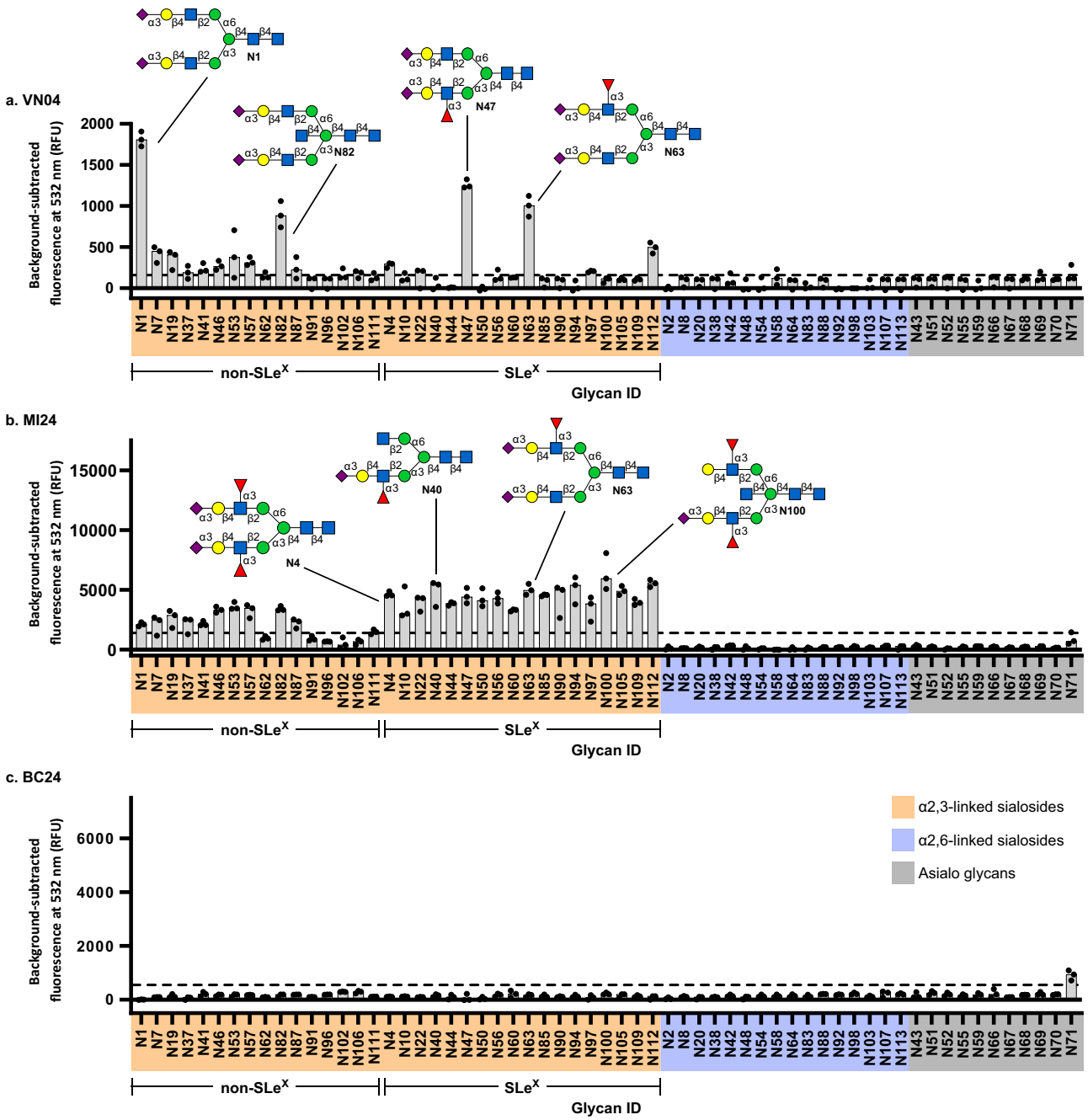



**Fig. 3 | *N*-linked glycan microarray analysis of H5 HA proteins showing loss of BC24 HA affinity for sialosides.** *N*-linked glycan binding profiles of HA proteins expressed from GnTI- Expi293F cells. **a** VN04 HA, **b** MI24 HA, and **c** BC24 HA. The horizontal dashed line spanning each chart represents a binding threshold of fifty-fold the background signal, and y-axes are scaled according to these thresholds. Each point is the signal from a technical replicate (separate printed spot on the same array; $n = 3$), and bar heights are median values. $SLe^X$, sialyl Lewis$^X$. Orange, α2,3-linked sialosides; blue, α2,6-linked sialosides; and grey, asialo glycans. Glycan structure schematics are prepared according to the Symbol Nomenclature for Glycans (SNFG)[58] and adapted with permission from Z Biotech[10]. A full list of glycan structures for each ID is available in Supplementary Fig. 10. Source data are provided as a Source Data file.

fluorescence microscopy (Fig. 6a). We then quantified the extent of fusion by probing the activity of reconstituted split NanoBiT luciferase (Fig. 6c). Statistically significant increases in luminescence relative to mock-transfected controls were observed in all HA-transfected cultures (Fig. 6c and Supplementary Table 2). The levels of fusion mediated by the BC24 HA were substantially lower than by VN04 and MI24 HA (Fig. 6c), with BC24 HA yielding 0.32-fold the signal of VN04 HA, and MI24 1.3-fold (mean values). Given the absence of detectable glycan binding by BC24 HA in the structural and glycan array binding experiments, it is possible that fusion events with BC24 HA (Fig. 6a, c) may be driven by avidity effects, compensating for the low biochemical affinity of individual interactions[25,26]. It should be noted that BC24 HA-mediated fusion appears to be slightly greater than that of the Y98F severely reduced receptor binding mutant[26–28] on VN04 HA background (0.32-fold and 0.11-fold compared to wild-type VN04, respectively; Supplementary Fig. 8).

MEDI8852 is a broadly neutralising antibody which recognises a site on HA that is conserved in most HA subtypes[29]. To test whether BC24 was susceptible to neutralisation by MEDI8852, we repeated fusion assays with a dilution series of MEDI8852 $F_{ab}$ prior to HA

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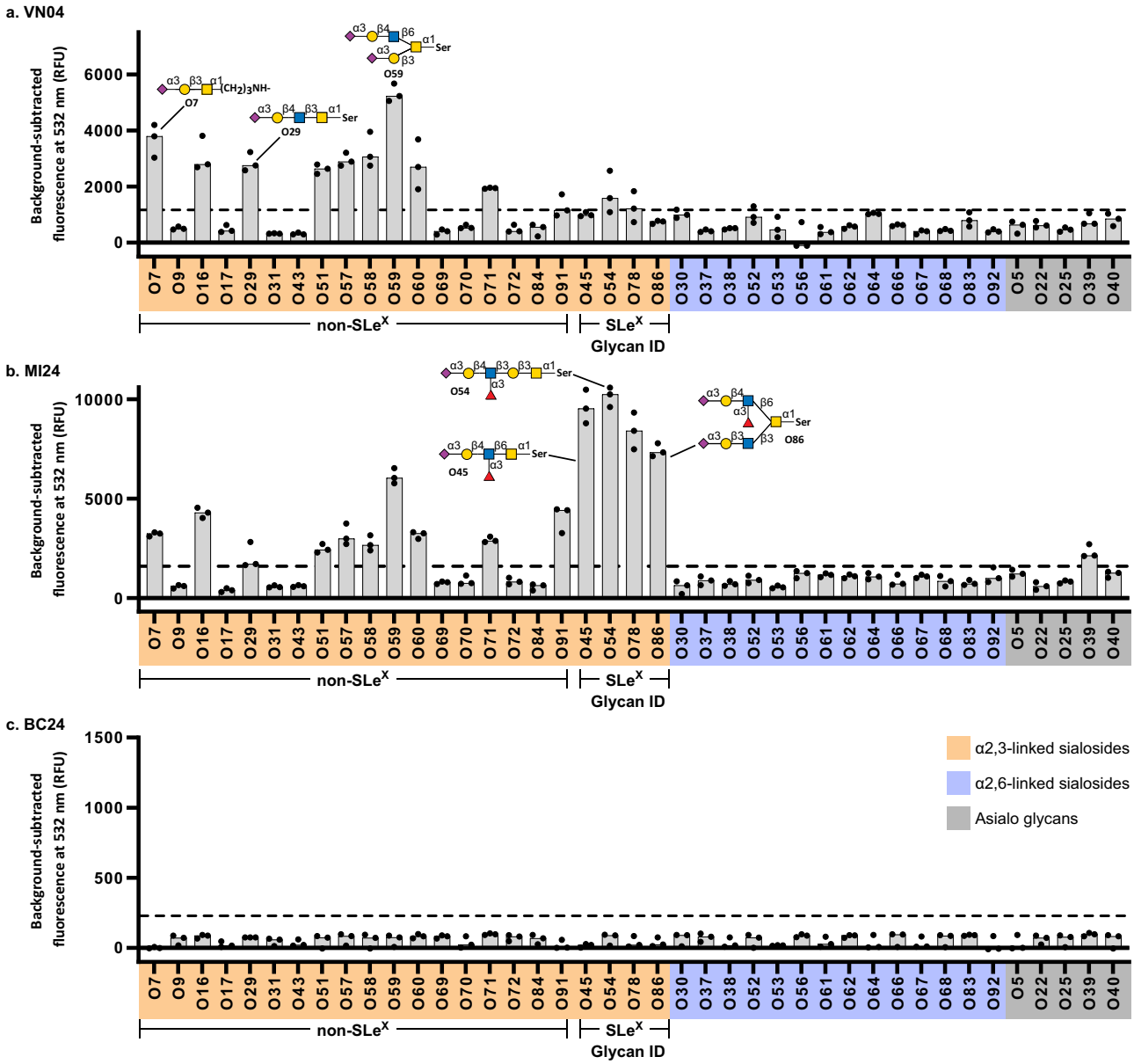



**Fig. 4 | *O*-linked glycan microarray analysis of H5 HA proteins showing loss of BC24 HA affinity for sialosides.** *O*-linked glycan binding profiles of HA proteins expressed from GnTI- Expi293F cells. **a** VN04 HA, **b** MI24 HA, and **c** BC24 HA. The horizontal dashed line spanning each chart represents a binding threshold of fifty-fold the background signal, and y-axes are scaled according to these thresholds. Each point is the signal from a technical replicate (separate printed spot on the same array; $n = 3$), and bar heights are median values. $SLe^X$, sialyl Lewis$^X$; RFU, relative fluorescence units. Orange, α2,3-linked sialosides; blue, α2,6-linked sialosides; and grey, asialo glycans. Glycan structure schematics are prepared according to the Symbol Nomenclature for Glycans (SNFG)[58] and adapted with permission from Z Biotech[11]. A full list of glycan structures for each ID is available in Supplementary Fig. 11. Source data are provided as a Source Data file.

activation to estimate the extent to which BC24 HA activity can be neutralised by this antibody (Fig. 6b, d). $IC_{50}$ values computed based on NanoBiT luciferase assays demonstrate that BC24 is potently neutralised by MEDI8852 at least as effectively as VN04 and MI24 (estimated $IC_{50}$ values of 25.3 nM and 31.9 nM for VN04 and MI24 HA, respectively). Biolayer-interferometry (BLI) experiments with MEDI8852 $F_{ab}$ validate specific binding in the nanomolar range (Supplementary Table 3 and Supplementary Fig. 9) with all three HA proteins.

## Humoral immune response against BC24 H5 HA

Based on our findings that influenza A which demonstrated fitness in the severe BC24 case had HA variants (up to a third of sequence reads carrying E190D or Q226H mutations[3]) whose binding to sialosides was undetectable by microarray (Figs. 3 and 4) and ELISA (Fig. 5) analysis, and that BC24 H5 HA caused fewer fusion events (Fig. 6c), we sought to determine the BC24 patient's antibody titre against BC24 H5 HA. We tested serum samples obtained from the BC24 patient at four independent time points within 1 to 4 days following negative serum Flu A RT-PCR test results[3] and from four age-matched healthy controls (Supplementary Table 4). Titration of serum antibodies against BC24 and MI24 HA proteins showed that healthy controls have modest reactivity against both of these clade 2.3.4.4b H5 HA proteins (Fig. 7a). Since these individuals have no known prior history of exposure to H5N1 influenza, these are likely cross-reactive antibodies generated through vaccination or in response to circulating strains of influenza. In contrast, sera from the BC24 patient demonstrate strong reactivity against both the BC24 and MI24 HA proteins (Fig. 7a), indicating that

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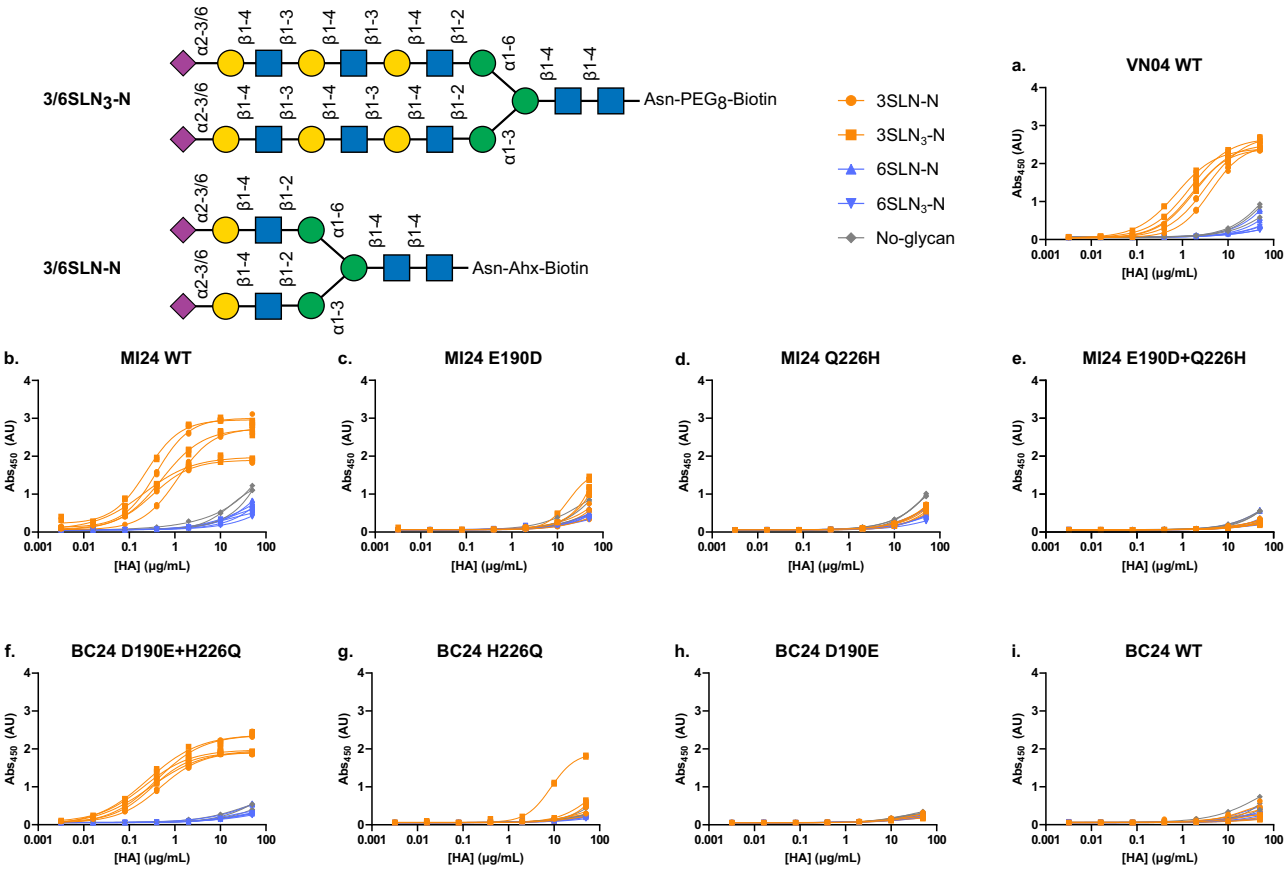



**Fig. 5 | Individual BC24 HA receptor-binding mutations reduce binding to α2,3- and α2,6-linked sialosides.** ELISA experiments probing HA binding to immobilised biantennary α2,3- (orange curves) and α2,6-linked (blue curves) sialosides (names and Symbol Nomenclature for Glycans (SNFG)[58] at top left; $PEG_8$, octaethylene glycol; Ahx, 6-aminohexanoic acid). **a** VN04 wild-type, **b** MI24 wild-type, **c** MI24 E190D mutant, **d** MI24 Q226H mutant, **e** MI24 E190D + Q226H double mutant, **f** BC24 D190E + H226Q variant, **g** BC24 H226Q variant, **h** BC24 D190E variant, and **i** BC24 wild-type. Each curve represents a separate biological replicate with independent preparations of HA ($n = 3$). Each point plotted is a technical replicate ($n = 3$) within each independent experiment. $Abs_{450}$, absorbance at 450 nm; AU, absorbance units. Source data are provided as a Source Data file.

the patient mounted a robust antibody response to the BC24 H5 HA protein.

To determine whether this reactivity was specific to H5 HA, we repeated the ELISA experiment against HA proteins from A/California/04/2009 (H1N1), A/Singapore/1/1957 (H2N2), and A/Perth/16/2009 (H3N2). As expected, the healthy controls showed a robust response against H1 and H3 HA proteins, which are part of circulating strains of influenza A and incorporated in seasonal vaccines, and mild reactivity to H2 HA (Fig. 7b–d). Notably, sera from the BC24 patient reacted strongly against all three HA proteins, including H2 HA even at the highest dilution (Fig. 7b–d) despite the negative nasopharyngeal swab obtained for A(H1) and A(H3) viral infection at the time of admission[3].

## Discussion

Our structural, biochemical, and cellular analyses collectively demonstrate severely diminished affinity of BC24 HA to α2,3-linked sialosides on host cells. The structures of several H5 HA proteins reveal a lack of BC24 HA binding to the auto-glycan (Fig. 1d). In contrast, the MI24 HA which differs from BC24 HA by only nine residues in the ectodomain per protomer (Supplementary Fig. 1), and by only E190D and Q226H amongst receptor-binding site residues, is competent to bind the auto-glycan, showing continuous glycan density even in the absence of stabilisation by antibodies[15]. Given the similarity in receptor-binding site conformations between the BC24, VN04, and MI24 HA proteins (Fig. 2g, h), the reduced auto-glycan affinity likely arises from a lack of specific interactions with the altered 190 and 226 positions, where E190 and Q226 appear to be favourable to binding in other strains.

Extending the analysis to α2,6-linked sialosides and addressing potential differences in auto-glycan binding compared to receptor binding, the glycan microarray and ELISA experiments further demonstrate that the BC24 HA detectably binds neither the α2,3-linked sialosides nor the α2,6-linked sialosides assayed (Figs. 3c, 4c, and 5i). Indeed, the mutagenesis experiments with ELISA-based readout suggest that each of the E190D and Q226H mutations yield reduced affinity for both types of sialosides in H5 HA (Fig. 5c–h). This isolation of the mutations allows us to comment on the lack of public information on the frequency at which the E190D and Q226H mutations reside on the same alleles[3]. At least in regard to binding to the commonly representative 3SLN-N, $3SLN_3$-N, 6SLN-N, and $6SLN_3$-N sialosides, each single mutant and the double mutant displays a similar reduced binding phenotype. Therefore, receptor binding of the mutant viruses found in this BC24 case does not appear to depend on the specific combination of E190D and Q226H mutations. As populations of single mutants are expected to have existed in the patient[3], it is interesting that multiple variants existed with this similar phenotype.

Our cell-based assay showing reduced fusion in the BC24 HA-transfected cultures when compared to VN04 and MI24 HA further supports the interpretation that affinity for the most common glycans is reduced (Fig. 6c). The A549 alveolar cells used in this assay have been previously demonstrated to express both α2,3- and α2,6-linked sialosides on their surface[30]. The BC24 HA may thus be binding biologically

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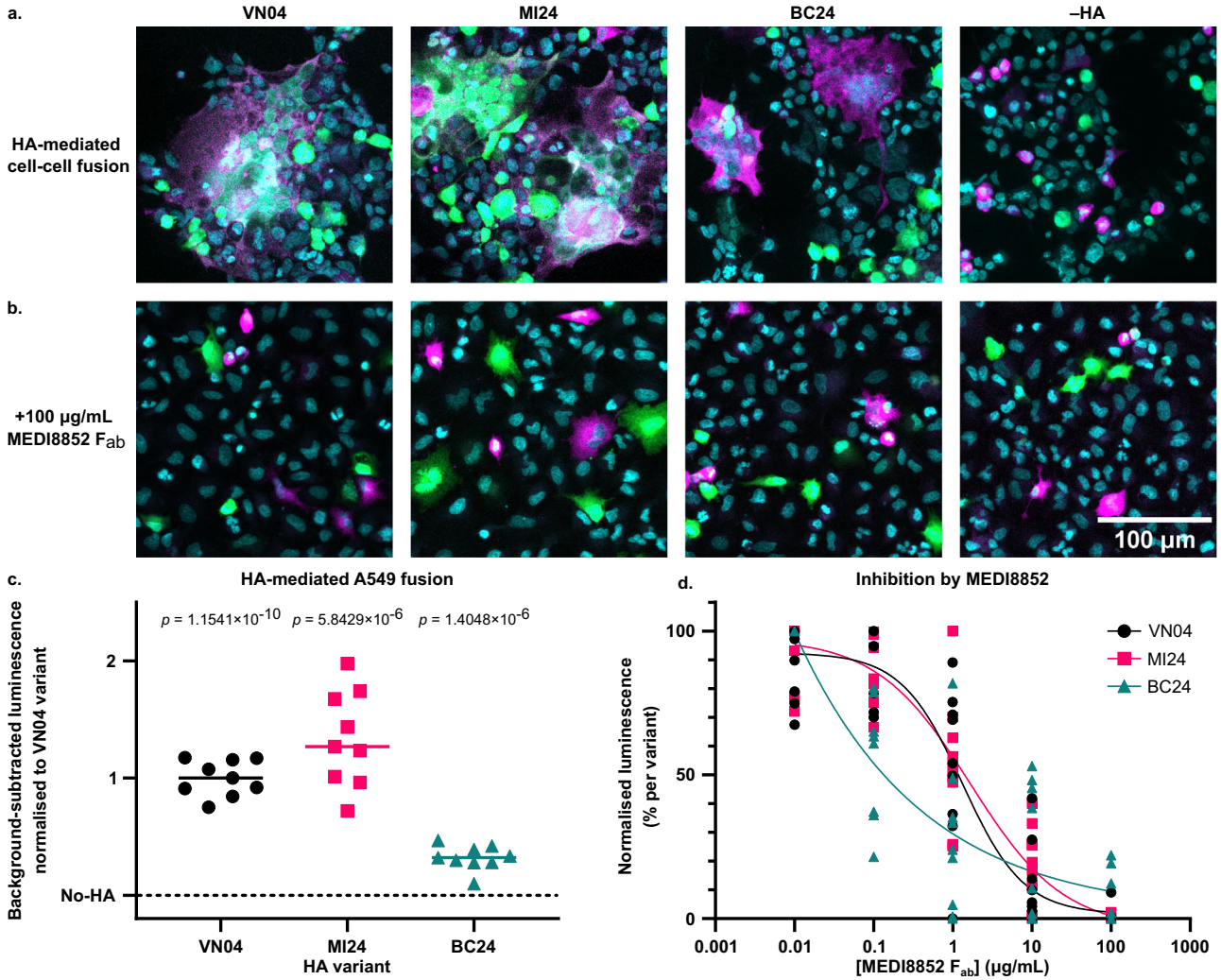



**Fig. 6 | BC24 HA is fusion competent and neutralizable by MEDI8852.** **a** Representative micrographs of A549 human alveolar carcinoma cells showing syncytia following acidifcation uniquely in HA-transfected conditions. Green, GFP; magenta, mCherry; and cyan, DAPI. **b**. Representative micrographs with inhibition of fusion by 1 h incubation with MEDI8852 $F_{ab}$ prior to acidification. **c** Summary of fusion efficiency by each HA determined by luciferase assay for NanoBiT reconstitution of separately transfected cells. Data presented have background signal from mock-transfected controls subtracted and are subsequently normalised to the signal from the VN04 HA within each set of biological replicates. Horizontal lines represent the median. Significance values are against mock-transfected control and determined by two-tailed t-test with Welch's correction, with parameters provided in Supplementary Table 2. **d** Dose-inhibition curves of MEDI8852 $F_{ab}$ for each HA with y-axis normalised to be the range of observed signal for each HA. **c, d** In both sets of experiments, three biological replicates (distinct culture passages) were performed, each including three technical replicates as separate wells on the culture plates. Source data are provided as a Source Data file.

relevant sialosides at low affinity below the limit of detection for ELISA and microarray analysis but partially compensated by avidity in the fusion assay[26], or binding to alternative receptors not assayed in this study. With the caveat that the A549 cell line does not perfectly imitate real lung tissue, a better understanding of influenza infectivity in spite of severely reduced viral entry as suggested by the fusion assays may be of interest in outbreak preparedness (Fig. 6c). Of note, our assay shows that BC24 HA is capable of mediating levels of fusion that are statistically significantly greater than the commonly employed Y98F mutant, although the ranges of signal overlap (Supplementary Fig. 8). Viral proliferation has been previously reported with this Y98F mutation, depending on cell type[26,28], suggesting that the reduced-binding minority variants of interest in the BC24 case are functional. Ultimately, our findings do not support the hypothesis[3] that the mutations at residues 190 and 226 may have led to a switch in HA receptor specificity from α2,3-linked to α2,6-linked sialosides.

Given that sequences were derived from a lower respiratory tract sample[3], we propose a hypothesis that the severe reduction in binding affinity, although to a level which allows fusion competence (Fig. 6c), may have enriched the distal airways in these BC24 influenza variants by helping to avoid receptors present in the upper respiratory tract. We lack the spatial and temporal sequencing information to comment further on viral population dynamics, and it is unknown whether the BC24 minority variants contributed to disease, did not affect symptom severity, or were themselves permitted by other factors. Nevertheless, infection of the lower respiratory tract, as was found in this case, promoted or not by the reduced binding phenotype, is associated with disease severity[3,31,32].

Interestingly, the HA protein (MI24) which demonstrated the strongest binding to sialosides in the microarray analysis (Figs. 3b and 4b) and mediated the highest levels of A549 cell fusion (Fig. 6c) was identified from the influenza case with the mildest symptoms, limited to conjunctivitis[33]. By contrast, the VN04 and BC24 cases were respectively fatal and severe[3,34], while demonstrating weaker receptor binding (Figs. 3a, c and 4a, c), although there are biologically relevant glycans not explored in this study[7,9]. Many complex factors are

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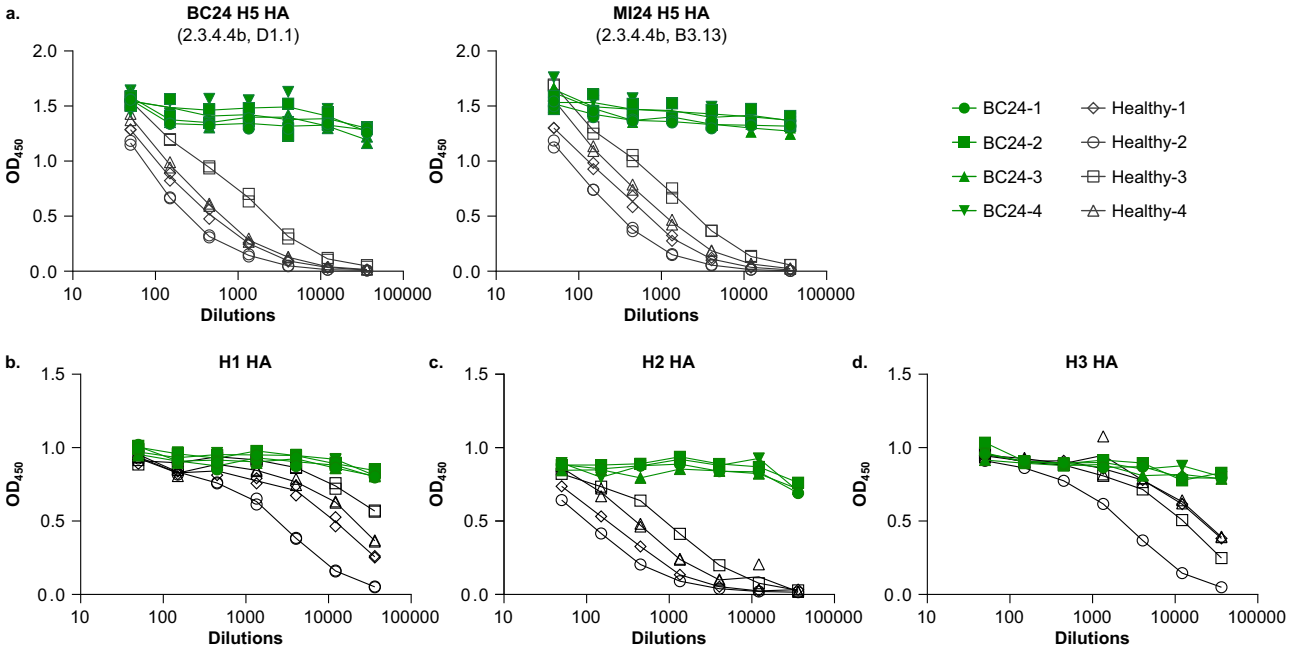



**Fig. 7 | Potent BC24 patient serum reactivity against multiple HA subtypes.** Binding signal for immobilised **a** BC24 (left) and MI24 (right) H5 HA, **b** H1 HA, **c** H2 HA, and **d** H3 HA proteins incubated with serially diluted serum samples (green, sera from the BC24 patient collected at four different time points immediately following a negative serum Flu A RT-PCR test result; white, sera from unremarkable age-matched healthy donors). The data represent background-subtracted measurements of optical density at 450 nm ($OD_{450}$) in ELISA experiments, with each point being a technical replicate ($n = 2$). Lines connect the mean signal at each serum dilution. Persistence of signal with increasing dilution indicates a greater titre of reactive antibodies. Source data are provided as a Source Data file.

involved in disease severity, but one potential contributor could be our hypothesised spatial distribution of these viral variants assisted by weak receptor binding. Indeed, the patient had positive Flu A RT-PCR results for a longer time for samples from tracheal aspirate compared to throat swabs, and higher viral loads in the lower respiratory tract as compared to the upper respiratory tract[3]. Investigation of the H5N1 outbreaks in the 2000s showed a pattern of symptoms predominantly in the lower respiratory tract, contributing to disease severity[31]. Similarly, Oner et al. had found that presentation with rhinorrhea correlated with milder disease, speculating on a protective effect by not permitting infection of the lower respiratory system[32]. In any case, it is interesting that despite not being the majority variants at the time sequencing samples were taken, the mutations were detected in approximately a third of reads in that region[3], suggesting that in this case weakly binding variants at least remained relatively fit.

The microarray used in this study does not include sulfated glycans which are displayed on respiratory cells[4]. These glycans have previously been shown to bind to H5 HA proteins, and the VN04 and MI24 HA proteins tended to bind more strongly to sulfated glycans than non-sulfated ones[9]. Interestingly, H1 HA proteins were found to bind sulfated α2,3-linked sialosides with both aspartate or glutamate at position 190 while the non-sulfated α2,3-linked sialoside binding was hindered by introducing the E190D mutation[7]. It is possible that BC24 may use other receptors than those tested in our microarray experiments[10,11] for host cell recognition.

The BC24 case did not result in any subsequent human-to-human transmission, which is consistent with the prevailing understanding that α2,6-linked sialoside binding promotes human spread[35,36]. However, this BC24 case demonstrates severe disease at least at the individual health level despite lacking human disease markers like increased α2,6-linked sialoside affinity in the HA protein. Previous examples in other respiratory viruses such as the 2015 outbreak of MERS-CoV suggest that receptor-binding strength is not necessarily a determinant of transmissibility[37]. Ultimately, the adaptation of viruses to human infectivity requires further study to unveil such nuances. Fusion activity-based screening may provide an additional important approach to better understand zoonotic potential in surveillance efforts.

Hemagglutinin is just one gene which must be compatible with human infection. Since functional balance must be maintained for hemagglutinin and the viral receptor-destroying enzyme neuraminidase, reduction in hemagglutinin binding ability to host receptors may be hypothesised to be accompanied by compensatory changes in neuraminidase function[38]. Genotype D1.1 H5N1 influenza like BC24 carry an N221S mutation (N1 numbering) at the neuraminidase active site, although whether this affects enzymatic activity is unknown[39]. The neuraminidase gene sequenced from the BC24 patient does not carry stalk length mutations which have been shown to modulate function[39]. Other factors, such as the relative proportion of hemagglutinin and neuraminidase molecules decorating the virion surface, are unknown. Further investigation into neuraminidase function in genotype D1.1 H5N1 human infections is needed.

Pending further investigation of the other genes sequenced from the BC24 strains, explanations for the BC24 case include host factors that are independent of the nature of HA. Patient age may have played a role, noting that teen-aged children have historically been particularly susceptible to severe disease attributed to H5N1 influenza due to lack of pre-existing immunity and different receptor expression profiles[4,32,40]. Factors such as asthma may also have played a role in the severity of the infection[3,41–43]. Although a specific understanding of the underlying causes differs between studies, with environmental factors often being substantial covariates, hyperresponsive immune systems are a consideration in the study of infections in patients with asthma[3,41–44].

Although BC24 influenza with the studied mutations represented a significant portion of virus in a severe disease case, both of the recent BC24 and MI24 HA proteins remain as readily neutralised by MEDI8852 as the VN04 HA (Fig. 6d). Most clade 2.3.4.4b H5N1 continue to be susceptible to commonly licensed anti-influenza drugs[45], and the neuraminidase gene sequenced from this patient does not carry the

H275Y (N1 numbering) mutation circulating in British Columbia around the time[46], nor any other known resistance mutations[45,47]. Patient samples demonstrated susceptibility to oseltamivir[3]. Concerns of near-term health emergencies from zoonotic transmissions of related flu strains are largely assuaged because of the lack of resistance of these viral strains to traditional antiviral methods, and because the HA itself retains the conserved stem region that is efficaciously targeted by broadly-neutralising antibodies (Fig. 6b,d and Supplementary Fig. 9).

The serum ELISA experiments demonstrated that all four healthy pediatric samples had detectable antibodies against clade 2.3.4.4b BC24 and MI24 HA proteins (Fig. 7a), likely due to cross-reactive antibodies from pre-existing immunity against vaccine or circulating strains of influenza A. Although these data are not sufficient to determine the capacity of the antibodies to neutralise the virus, they support growing evidence of cross-reactivity against H5 HA following influenza vaccinations or infection by other subtypes, indicating that the general population may not be entirely naïve to H5N1 influenza in the case of zoonotic events[48–50]. Similarly, we showed that pre-existing antibodies cross-react with HA from a non-circulating H2N2 strain as well (Fig. 7c).

Serum samples from the BC24 patient demonstrated potent antibody reactivity against the BC24 and MI24 HA proteins (Fig. 7a) and strong cross-reactivity to H1, H2, and H3 HA proteins (Fig. 7b–d). The heightened response to HA is consistent with the timing of the serum collection being at the peak of a typical antibody response to infection. This result suggests that the patient was able to mount a strong humoral response against this BC24 influenza, at least to the HA protein, yet this antibody response was insufficient to mitigate severe disease. In conjunction with evidence of hypercytokinemia[3], it is also possible that antibody-dependent enhancement contributed to disease severity, but this requires further investigation. Nevertheless, our findings show that BC24 HA does not escape targeting by the humoral immune response.

The cryo-EM structural analyses demonstrate that the combined effect of the E190D and Q226H mutations sequenced from BC24 HA alters the glycan-binding behavior of this HA. Subsequent experiments through glycan microarray analyses, ELISA, and fusion assays in A549 cells support the hypothesis that the BC24 HA has severely reduced affinity for both α2,3-linked and α2,6-linked sialosides. In particular, ELISA experiments show that both the E190D and Q226H mutations reduce sialoside affinity in isolation, as well. Therefore, multiple distinct minority variants in the BC24 case are expected to have a reduced binding phenotype. We show that the virus likely retains the ability to drive membrane fusion, adding complexity to our understanding of avian influenza's paths to generating severe disease in humans. Several important conclusions are suggested by our analyses. The lack of binding to α2,6-linked sialosides by HA variants in the BC24 case implies that human infection is not always dependent on such binding. Moreover, generally reduced binding of HA to sialosides, including both α2,3- and α2,6-linked sialosides, can be a phenotype of fit viruses in a severe human disease context. We speculate that such a reduction in binding may enable deeper penetration into the lung and potentially contribute to a more severe infection; however, further study will better elucidate how differences in receptor-binding sites influence the infectivity of this influenza strain.

## Methods

### Construct design and mutagenesis

VN04 H5 HA: GenBank AAW80717.1; MI24 H5 HA: GenBank XBE32674.1; BC24 H5 HA: GenBank XTJ61994.1; H1 HA: ACP41105.1; H2 HA: AAA43678.1; H3 HA: ACS71642.1

Given the mixed population of HA sequences identified from the patient[3], the BC24 HA studied here carries both the E190D and Q226H mutations that were most frequently sequenced from patient samples.

**Table 1 | Primer sequences for site-directed mutagenesis**

| Gene | Mutation | Primer Sequences (5′ → 3′) |
|---|---|---|
| MI24 HA ectodomain | E190D | Forward: CAATGCCGAGGATCAGACAAATC<br>Reverse: TTGCTGTGGTGGATG |
| | Q226H | Forward: GGTGAACGGACATAGAGGCAGAA<br>Reverse: TGGCTTCTGGTTGCG |
| BC24 HA ectodomain | D190E | Forward: GTA-CAGGTTGGTCTGCTCCTCGGCGTTGTTG<br>Reverse: CAACAACGCCGAGGAGCAGACCAACCTGTAC |
| | Q226H | Forward: AGTGAACGGACAGAGAGGCAGAATG<br>Reverse: TGGGACCGTGTGGCG |
| VN04 HA full-length | Y98F | Forward: TCGCCGGGGAAGCA-CAGGTCGTTCACA<br>Reverse: TGTGAACGACCTGTGCTTCCCCGGCGA |

For structural analysis, we designed soluble HA ectodomain constructs (residues 1-521, H5 sequential numbering) with M36C and G392C (H5 sequential numbering) disulphide-stabilisation mutations[51] and C-terminal AviTag, thrombin cleavage site, foldon trimerization domain, and 8xHis tag. Supplementary Fig. 1 shows a sequence alignment of these ectodomain constructs prepared using Clustal Omega and visualised using MView made available through EMBL-EBI resources[52]. Since the stem region epitope is distal to the receptor-binding site, the BLI experiments with MEDI8852 were performed using HA carrying an additional Y107F[27] (H5 sequential numbering) mutation to reduce receptor binding to improve yields and to reduce heterogeneous signal. We expressed these HA with their natural signal peptide sequences from a pcDNA3.1(+) backbone. Plasmids were codon optimised and synthesised at GenScript (Piscataway, NJ, USA). We performed site-directed mutagenesis (primers synthesised at Thermo Fisher Scientific, and primer sequences provided in Table 1) for the receptor-binding site mutations using the Q5 Site-Directed Mutagenesis kit (New England Biolabs) with validation by whole-plasmid sequencing at Plasmidsaurus (South San Francisco, CA, USA). For fusion assays, we expressed full-length wild-type untagged HA from pcDNA3.1(+) using the natural signal peptide sequences. Plasmids were obtained by custom synthesis at GenScript (Piscataway, NJ, USA).

### HA soluble ectodomain expression and purification

We expressed HA ectodomains in Expi293F cells (Gibco A14527) for cryo-EM, BLI, and serum ELISA experiments and in GnTI- Expi293F cells (Gibco A39240) for microarray analysis, glycan-binding ELISA experiments, and DLS. We transfected cells at $3\times10^6$ cells/mL by complexing HA-encoding plasmids (1.5 µg/mL of culture) with threefold mass excess of linear polyethyleneimine diluted in Expi293 Expression Medium (Gibco). Cells were subsequently incubated at 37°C in 8% CO2 at 130rpm (25mm orbital radius) with the addition of 2.2 mM valproic acid (100X in phosphate-buffered saline (PBS), pH 7.4) 1 day post-transfection. 5 days post-transfection, we harvested media supernatants containing HA ectodomain by centrifugation at 500 × *g* then clarified them by centrifugation at 15 000×g and filtration through a 0.45 µm pore size PES membrane.

We purified HA from the clarified supernatants on an ÄKTA pure system (GE Healthcare) at 4 °C by nickel affinity chromatography (loading onto a 5 mL HisTrap HP (Cytiva); washing with 20 mM Tris pH 8 500 mM NaCl, 20 mM imidazole; and elution in 20 mM Tris pH 8, 500 mM NaCl, 500 mM imidazole). For microarray, glycan-binding ELISA, and DLS experiments, we buffer exchanged affinity-purified product with Amicon Ultra centrifugal filters (Millipore), 100 kDa molecular-weight cutoff (4 mL) to TBS (20 mM Tris pH 8, 150 mM NaCl). For cryo-EM, BLI, DLS (where specified), and serum ELISA experiments, we further purified samples by size-exclusion

chromatography (SEC; Superdex 200 Increase 30/100 GL (Cytiva), eluted with TBS) before concentration by centrifugal filtration to 1.7-2.0 mg/mL.

### Cryo-electron microscopy

We applied 1.8 µL of purified HA ectodomains at 10 °C, >98% humidity onto Quantifoil R1.2/1.3, 200 mesh, Cu holey carbon grids glow discharged for 20 s using a Pelco easiGlow. Using a Vitrobot Mark IV (Thermo Fisher Scientific), we then blotted grids for 12 s at −10 relative blot force and plunge-froze samples in liquid ethane. We collected micrographs using a Titan Krios G4 transmission electron microscope (Thermo Fisher Scientific) at 300 kV with a Selectris X imaging filter set to 10 eV slit width and Falcon 4 direct electron detector in electron event registration mode. We collected movies at 165 000x magnification over a defocus range of −1.0 to −2.0 µm with a total dose of 40.0 $e^-/Å^2$.

We performed data processing using CryoSPARC v4.6.2, with the workflow provided in Supplementary Figs. 2–4. In brief, following patch motion correction and CTF estimation, we performed blob particle picking. After iterative rounds of heterogeneous refinement, we generated the final reconstruction with non-uniform refinement with C3 symmetry imposed. We fit an initial model of a $F_{ab}$-complexed VN04 HA ectodomain crystal structure into the maps using UCSF ChimeraX v1.7[53]. We then subjected the model to several cycles of real-space refinement in Phenix v1.21.1 and manual adjustment on WinCoot v0.9.8[54,55]. Model statistics are available in Supplementary Table 1. We introduced mutations as needed using WinCoot. Figures were prepared using UCSF ChimeraX. RMSD values were computed on PyMOL (Schrödinger).

### MEDI8852 $F_{ab}$ preparation

We expressed MEDI8852 $F_{ab}$ (UniRef100_UPI0008141F01, UniRef100_UPI0008141F29) from pcDNA3.1(+) backbones, with the heavy chain carrying a C-terminal 6xHis tag. Plasmids were synthesised by GenScript. We expressed MEDI8852 $F_{ab}$ in Expi293F cells in the same manner as the soluble HA ectodomains, transfecting with 1 µg each of heavy- and light-chain-encoding plasmid per mL of culture. We purified $F_{ab}$s by nickel affinity chromatography and size-exclusion chromatography in the same manner as the soluble HA ectodomains.

### Glycan microarray experiments

Microarray experiments using the Z Biotech N-glycan (10602-16 K; Supplementary Fig. 10) and O-glycan (10611-16 K; Supplementary Fig. 11) array kits[10,11] and subsequent signal processing were performed by Creative Proteomics (Shirley, NY, USA). His-tagged HA ectodomains (20 µg/mL) were precomplexed in glycan array assay buffer (GAAB; Z Biotech) at a 10:1:1 mass ratio with rabbit anti-6xHis antibody (Invitrogen PA1-983B) and Cy3-conjugated anti-rabbit IgG (Life Technologies A10520) for 1 h on ice. The microarray was blocked for 30 min at room temperature in glycan array blocking buffer (Z Biotech). 100 µL of precomplex HA-antibodies solution was added to each subarray and incubated for 1 h at room temperature prior to washing in GAAB then water pursuant to manufacturer instructions. The microarray slide was then scanned for fluorescent signal at 532 nm in high laser power mode on an InnoScan 710 microarray scanner (Innopsys), and the data were processed on Mapix (Innopsys) software to subtract background signal from glycan-free spots. Glycan schematics adapted from Z Biotech for Figs. 3 and 4 and produced on GlycoGlyph with adjustments on Inkscape for Fig. 5[10,11,56].

### Dynamic light scattering

We analysed using a Prometheus Panta (NanoTemper) the protein preparations used for glycan microarray analyses as well as positive and negative comparisons. For a more purified preparation, BC24 HA ectodomain from GnTI- Expi293F cells, further purified by SEC, was included. For an aggregated comparison, we subjected a portion of the BC24 HA sample (not purified by SEC) to four rounds of freezing and thawing. 0.2 mg/mL HA ectodomains were assayed on a Prometheus Panta in technical triplicate. The Size Analysis dynamic light scattering (DLS) procedure was run in high sensitivity mode at 25 °C. Data were processed automatically by Panta software, and figures were generated with the same software.

### Glycan-binding ELISA experiments

We immobilised biotin-conjugated biantennary glycans (Sussex Research BT000040, BT000055, BT000050, and BT000060) at 2.4 µM on Reacti-Bind streptavidin-coated 384-well plates (Pierce) at 4 °C for 16 h in PBS pH 7.4. Wells were then blocked in 6 mM d-desthiobiotin in PBS pH 7.4 for 1.5 h at room temperature. We concentrated preparations of soluble HA ectodomains to 1.5-3.5 mg/mL prior to precomplexing for 1 h at room temperature with peroxidase-conjugated rabbit anti-His tag IgG (Jackson ImmunoResearch 300-035-240) at a 2:1 molar ratio of HA trimer to antibody to improve binding signal[57]. We prepared this precomplex solution in TBS supplemented with 1% (w/v) bovine serum albumin (BSA). We then added HA-antibody complexes at the desired dilutions in BSA-supplemented TBS to the plate for incubation at room temperature for 2 h. Following washing five times in PBS pH 7.4 with 0.05% Tween-20 (PBS-T), we developed a signal by the addition of Pierce 1-Step Turbo TMB-ELISA substrate for 13 min at room temperature before quenching by the addition of an equal volume of 2M sulphuric acid. We then immediately measured absorbance at 450 nm using a Varioskan LUX (Thermo Scientific). We used GraphPad Prism v10.6.0 for all data analysis and binding curve generation. Using the Sigmoidal dose-response (variable slope) method, we fit curves for data with $\log_{10}$-transformed concentration values. We produced glycan schematics using GlycoGlyph then adjusted them on Inkscape[56].

### Cell-based HA fusion assay

We seeded $2 \times 10^4$ A549 cells (ATCC CCL-185) onto transparent-bottomed, black-walled 96-well plates (Greiner Bio-One) in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% heat-inactivated foetal bovine serum and penicillin-streptomycin. We incubated cultures at 37 °C in 5% $CO_2$. 16 h post-seeding, we transfected cells following the Lipofectamine 3000 protocol (Invitrogen). For each technical replicate, we transfected one group of cells to express GFP-HiBiT (Addgene 162589), and the other to express both the full-length HA of interest and mCherry-lgBiT (Addgene 199715)[23,24]. After an additional 24 h, we detached HA-transfected cells by 8 min incubation in 0.025% trypsin-EDTA at 37 °C and transferred them onto the GFP-HiBiT-transfected cells to incubate for 24 h. We activated HA by replacing the media with 90 mM sodium citrate pH 4.5, 150 mM NaCl, prewarmed to 37 °C, for 3 min, then immediately washed the cells in supplemented DMEM and let them recover in a second change of media for 6 h. We measured reconstituted NanoBiT activity as a proxy of fusion using the Nano-Glo Live Cell Assay System (Promega), recording luminescence with 1 s integration time 5 min after addition of substrate using a Varioskan LUX at 37 °C. We determined significance values by two-tailed t-test with Welch's correction on Microsoft Excel. 95% confidence intervals and $\eta^2$-values were determined using the additional analyses for Welch's t-test on GraphPad Prism 10. Statistics parameters are provided in Supplementary Table 2.

We immediately fixed cells by incubation for 20 min in 4% paraformaldehyde (Electron Microscopy Sciences) diluted in PBS pH 7.4, with a wash in PBS both before and after fixation. Following incubation in 1 µg/mL 4′,6-diamidino-2-phenylindole (DAPI) for 10 min, we washed wells five times in PBS, then stored them in ProLong Glass Antifade (Invitrogen) before imaging. For experiments comparing VN04, MI24, and BC24 HA, we imaged cultures using a Zeiss LSM 880 confocal microscope. We acquired a z-stack of five 2 µm-thick optical sections,

and 2D projections are presented. For experiments comparing wild-type VN04 and Y98F VN04 HA, we imaged cultures using an Echo Revolve microscope with digital haze reduction.

We performed inhibition assays in the same manner except supplementing media with the indicated concentration of MEDI8852 $F_{ab}$ 1 h prior to acidification, during which time cells remained at 37 °C with 5% $CO_2$. We performed data analysis on GraphPad Prism 10 in the same manner as for glycan-binding ELISA experiments, except that data were first normalized within the range of signal for each technical replicate before merging and calculating $IC_{50}$ values.

### Western blotting to detect HA expression

We harvested fusion assay A549 cells co-transfected with mCherry-lgBiT and the indicated H5 HA (Supplementary Fig. 7) by addition of 0.2 M Tris pH 6.5, 8% (w/v) sodium dodecyl sulphate (SDS), 4.3 M glycerol, 6 mM bromophenol blue, 0.4 M dithiothreitol. We then heated harvested samples at 95 °C for 10 min and loaded one third of the sample onto a 4-20% gradient Mini-PROTEAN TGX gel (Bio-Rad) to be run at 140 V submerged in Tris/glycine/SDS running buffer (Bio-Rad). Following blotting onto a nitrocellulose membrane at 100 V for 75 min in 4 °C 25 mM Tris, 192 mM glycine, 10% (v/v) methanol, we validated transfer using Ponceau S. We washed the membrane several times in water, then blocked it for 10 min at room temperature in EveryBlot Blocking Buffer (Bio-Rad). We incubated the blot in 1:2000 rabbit anti-HA mAb (Sino Biological 86001-RM01) and 1:1500 BA3R mouse anti-β-actin loading control mAb (Invitrogen) diluted in EveryBlot Blocking Buffer for 18 h at 4 °C. Following washing three times for ten minutes in PBS-T, we then incubated the blot for 1 h at room temperature in 1:2000 peroxidase-conjugated anti-mouse IgG (Jackson ImmunoResearch 715-035-150) and 1:2000 anti-rabbit IgG antibodies (Promega W401B). After washing five times for ten minutes each in PBS-T, we developed signal using Clarity Western ECL Substrate (Bio-Rad) and imaged the blot on a ChemiDoc MP imaging system (Bio-Rad) with both colorimetric (to validate band sizes against the stained protein standards) and chemiluminescent channels.

### Biolayer interferometry to assay MEDI8852 $F_{ab}$:HA binding

We performed all experiments on a Gator Prime system (Gator Bio) set to 30 °C and in BLI buffer (TBS supplemented with 0.02% Tween-20). We equilibrated anti-human FAB probes (Gator Bio) for 600 s at 400 rpm in BLI buffer before a 120 s baseline step at 1000 rpm. We loaded MEDI8852 $F_{ab}$ diluted to 100 nM onto probes for 40 s with 400 rpm shaking, reaching ~1 nm loading shift, followed by another 120 s baseline at 1000 rpm. We carried out both the HA ectodomain association and dissociation steps for 1200 s at 1000 rpm. HA:HA interactions may result in inflated $K_D$ especially for the VN04 and MI24 HA proteins which have higher glycan affinities, so we sought to limit this effect by using Y98F HA mutants with reduced sialoside affinities[27].

We used GatorOne software (Gator Bio) for data processing and analysis. Processing included y-axis alignment of the association phase, inter-step correction around the dissociation phase, and double reference subtraction. Processed sensograms are presented with Savitzky-Golay filtering. We determined kinetics and affinity parameters presented in Supplementary Table 3 by fitting curves using a 2:1 global binding model with $R_{max}$ considered unlinked. We recreated sensograms using the seaborn version 0.13.0 library for Python.

### Patient sample collection

Sera from the BC24 patient and healthy age-matched controls were obtained at the British Columbia Children's Hospital and stored at their BioBank according to ethics approval from the UBC C&W Research Ethics Board (protocol H23-02299). Use of the collected samples for research was approved by the UBC Clinical Research Ethics Board (H24-03937).

### ELISA to assay serum reactivity to clade 2.3.4.4b HA

BC24 and MI24 HA ectodomains were coated on a 96-well Nunc MaxiSorp plate (Thermo Scientific) by incubation of 2 µg/mL HA overnight at 4 °C. Wells were subsequently washed three times in PBS-T before blocking for 2 h at room temperature in PBS supplemented with 1% bovine serum albumin (blocking buffer). Wells were then incubated overnight at 4 °C in serum samples serially diluted in blocking buffer, washed, and subsequently incubated in 1:5000 peroxidase-conjugated goat anti-human IgG (Jackson ImmunoResearch) for 1 h at room temperature before washing three times in PBS-T. We developed the plates with TMB substrate and measured the signal at OD450nm.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

The cryo-EM data generated in this study have been deposited in the Protein Data Bank under accession codes 10OV, 10OW, and 10OX and Electron Microscopy Data Bank under accession codes EMD-75352, EMD-75353, and EMD-75354. All other data that support this study are available within the paper and its supplementary information files. Source data are provided with this paper.

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## Author contributions

J.H.N., A.M.B., P.A., S.M.Z., G.R.O., and S.S. conceptualised the study direction. J.H.N. performed site-directed mutagenesis, protein production and purification, cryo-EM data processing, structure modelling, DLS experiments, glycan-binding ELISA experiments, cell-based fusion assays, western blotting, and BLI; analysed the data; and prepared the manuscript. S.M.Z. performed protein purification and DLS experiments. A.M.B. collected cryo-EM data. R.L. performed serum screening ELISA experiments. X.Z. performed cryo-EM data processing and structure modelling. K.S.T. prepared samples for cryo-EM and screened grids. M.T. supervised patient sample acquisition and subsequent experiments. S.S. supervised the project and prepared the manuscript. All authors reviewed the results and discussed their interpretation.

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## Competing interests

S.S. is the founder and CEO of Gandeeva Therapeutics Inc. The remaining authors declare no competing interests.

## Additional information

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